

scGen Human Primary-Use Consensus and Validation

Scope

This document records the primary-use consensus for the 50-paper blinded scGen human-validation sample. It does not replace the reviewers' complete six-field records.

Consensus rules

Forty-four labels were identical after normalization of capitalization and wording. Four disagreements were resolved by user adjudication: SCGEN-HV-024 = Mention only; SCGEN-HV-029 = Benchmark; SCGEN-HV-043 = Mention only; SCGEN-HV-050 = Benchmark. Reviewer 2 left the primary-use field blank for SCGEN-HV-037 and SCGEN-HV-040, so the completed Reviewer 1 labels were carried forward and marked as such.

Source records

Reviewer 1: reviewer_1_blind_form_simple (document ID 1Uxvd-IVbj3A-aVsxgiiLVFk_JgUz-thnRbQvb8tadPA)

Reviewer 2: reviewer_2_blind_form_simple (1) (document ID 1FuonfE2R-wGm0EAB-uctPywg1jKtTgGUIQBtaPpyDrM)

Consensus distribution

Mention only: 37; Benchmark: 9; Apply to biological data: 3; Other: 1; Extend or develop: 0. Total: 50.

Validation status

Primary-use performance was evaluated on the 40 preselected CLASSIFICATION cases. The 10 PROCESS_AUDIT cases were excluded because they test citation-retrieval and coverage decisions rather than citation-use classification. Agent predictions were taken from the final primary_label field in 20260903_scgen_rescue_consolidated_v1 and compared with the human consensus labels. Exact agreement was 37/40 (92.5%).

Primary-use precision and recall

Category	Human positives	Agent positives	TP / FP / FN	Precision	Recall
Apply to biological data	3	3	3 / 0 / 0	100.0%	100.0%
Extend or develop	0	2	0 / 2 / 0	0.0%	Not estimable
Benchmark	8	6	6 / 0 / 2	100.0%	75.0%
Mention only	28	29	28 / 1 / 0	96.6%	100.0%
Other	1	0	0 / 0 / 1	Not estimable	0.0%

Macro precision: 74.1%; macro recall: 68.8%. These are unweighted means of one-vs-rest category scores. A category-specific rate is not estimable when its denominator is zero; therefore the macro precision and macro recall each average the four estimable category rates. The three primary-use mismatches were SCGEN-HV-039 (agent Extend or develop; human Benchmark), SCGEN-HV-040 (agent Mention only; human Other), and SCGEN-HV-050 (agent Extend or develop; human Benchmark).

Case-level consensus

Case ID	Article	Independent labels	Consensus
SCGEN-HV-001	Opportunities for Artificial Intelligence in Advancing Precision Medicine DOI: 10.1007/s40142-019-00177-4	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-002	Species-agnostic transfer learning for cross-species transcriptomics data integration without gene orthology DOI: 10.1093/bib/bbae004	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-003	Modelling stem cell differentiation related processes—A practical overview for biologists DOI: 10.1002/1873-3468.70395	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-004	scDREAMER for atlas-level integration of single-cell datasets using deep generative model paired with adversarial classifier DOI: 10.1038/s41467-023-43590-8	Reviewer 1: Benchmark Reviewer 2: Benchmark	Benchmark Reviewer agreement
SCGEN-HV-005	Benchmarking LLM-based agents for single-cell omics analysis DOI: 10.1186/s13059-026-03998-z	Reviewer 1: Benchmark Reviewer 2: Benchmark	Benchmark Reviewer agreement
SCGEN-HV-006	BuDDI: Bulk Deconvolution with Domain Invariance to predict cell-type-specific perturbations from bulk DOI: 10.1371/journal.pcbi.1012742	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-007	Discovering governing equations of biological systems through representation learning and sparse model discovery DOI: 10.1093/nargab/lqaf048	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement

Case ID	Article	Independent labels	Consensus
SCGEN-HV-008	Disrupted developmental signaling induces novel transcriptional states DOI: 10.1073/pnas.2418351122	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-009	PerTurboAgent: A Self-Planning Agent for Boosting Sequential Perturb-seq Experiments DOI: 10.1101/2025.05.25.656020	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-010	Learning Perturbation-specific Cell Representations for Prediction of Transcriptional Response across Cellular Contexts DOI: 10.1101/2023.03.20.533433	Reviewer 1: Benchmark Reviewer 2: Benchmark	Benchmark Reviewer agreement
SCGEN-HV-011	A Cross-Modal Autoencoder Framework Learns Holistic Representations of Cardiovascular State DOI: 10.1101/2022.05.26.493497	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-012	scVAE: variational auto-encoders for single-cell gene expression data. DOI: 10.1093/bioinformatics/btaa293	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-013	Biologically Interpretable VAE with Supervision for Transcriptomics Data Under Ordinal Perturbations DOI: 10.1101/2024.03.28.587231	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-014	Fluctuation structure predicts genome-wide perturbation outcomes DOI: 10.1101/2025.06.27.661814	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-015	A practical guide to sequencing in neuropsychiatric research DOI: 10.1038/s44277-025-00041-0	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-016	Neural network-assisted humanisation of COVID-19 hamster transcriptomic data reveals matching severity states in human disease DOI: 10.1016/j.ebiom.2024.105312	Reviewer 1: Apply to biological data Reviewer 2: Apply to biological data	Apply to biological data Reviewer agreement
SCGEN-HV-017	Cells of the adult human heart DOI: 10.1038/s41586-020-2797-4	Reviewer 1: Apply to biological data Reviewer 2: Apply to biological data	Apply to biological data Reviewer agreement
SCGEN-HV-018	Causal identification of single-cell experimental perturbation effects with CINEMA-OT DOI: 10.1038/s41592-023-02040-5	Reviewer 1: Benchmark Reviewer 2: Benchmark	Benchmark Reviewer agreement
SCGEN-HV-019	Shortcomings of silhouette in single-cell integration benchmarking DOI: 10.1038/s41587-025-02743-4	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-020	Domain adaptation for supervised integration of scRNA-seq data DOI: 10.1038/s42003-023-04668-7	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-021	Combinatorial prediction of therapeutic perturbations using causally inspired neural networks DOI: 10.1038/s41551-025-01481-x	Reviewer 1: Benchmark Reviewer 2: Benchmark	Benchmark Reviewer agreement
SCGEN-HV-022	Application and prospects of single-cell and spatial omics technologies in woody plants DOI: 10.48130/fr-2023-0027	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-023	Can Pharmaceutical World Models Become Scientific Instruments? A State-of-the-Art Review of Simulation, Reasoning, and Validation DOI: 10.51847/hn9ejvjqoc	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-024	Assessing and mitigating batch effects in large-scale omics studies DOI: 10.1186/s13059-024-03401-9	Reviewer 1: Mention only Reviewer 2: Benchmark	Mention only User-adjudicated consensus
SCGEN-HV-025	VI-VS: calibrated identification of feature dependencies in single-cell multiomics DOI: 10.1186/s13059-024-03419-z	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement

Case ID	Article	Independent labels	Consensus
SCGEN-HV-026	The Power of Two: integrating deep diffusion models and variational autoencoders for single-cell transcriptomics analysis DOI: 10.1101/2023.04.13.536789	Reviewer 1: Benchmark Reviewer 2: Benchmark	Benchmark Reviewer agreement
SCGEN-HV-027	Single-cell RNA-seq reveals spatially restricted multicellular fibrotic niches during lung fibrosis DOI: 10.1101/569855	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-028	Causal Imputation via Synthetic Interventions DOI: 10.48550/arxiv.2011.03127	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-029	Style transfer with variational autoencoders is a promising approach to RNA-Seq data harmonization and analysis DOI: 10.1093/bioinformatics/btaa624	Reviewer 1: Mention only Reviewer 2: Benchmark	Benchmark User-adjudicated consensus
SCGEN-HV-030	Reconstructing unobserved cellular states from paired single-cell lineage tracing and transcriptomics data DOI: 10.1101/2021.05.28.446021	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-031	Learning genetic perturbation effects with variational causal inference DOI: 10.1371/journal.pcbi.1013194	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-032	Spatial single-cell mass spectrometry defines zonation of the hepatocyte proteome DOI: 10.1038/s41592-023-02007-6	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-033	Learning interpretable cellular and gene signature embeddings from single-cell transcriptomic data DOI: 10.1038/s41467-021-25534-2	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-034	Detecting subtle transcriptomic perturbations induced by lncRNAs knock-down in single-cell CRISPRi screening using a new sparse supervised autoencoder neural network DOI: 10.3389/fbinf.2024.1340339	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-035	lute: estimating the cell composition of heterogeneous tissue with varying cell sizes using gene expression DOI: 10.1186/s12864-025-11508-x	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-036	Learning cell communication from spatial graphs of cells DOI: 10.1101/2021.07.11.451750	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-037	Variational autoencoders learn universal latent representations of metabolomics data DOI: 10.1101/2021.01.14.426721	Reviewer 1: Mention only Reviewer 2: Blank	Mention only Reviewer 1 carry-forward; Reviewer 2 blank
SCGEN-HV-038	The manatee variational autoencoder model for predicting gene expression alterations caused by transcription factor perturbations DOI: 10.1038/s41598-024-62620-z	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-039	A deep generative model for deciphering cellular dynamics and in silico drug discovery in complex diseases DOI: 10.1038/s41551-025-01423-7	Reviewer 1: Benchmark Reviewer 2: Benchmark	Benchmark Reviewer agreement
SCGEN-HV-040	Differences between in vivo and ex vivo models drive divergent perturbation outcomes DOI: 10.1101/2025.09.02.673692	Reviewer 1: Other Reviewer 2: Blank	Other Reviewer 1 carry-forward; Reviewer 2 blank
SCGEN-HV-041	A Roadmap for a Consensus Human Skin Cell Atlas and Single-Cell Data Standardization DOI: 10.1016/j.jid.2023.03.1679	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement

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SCGEN-HV-042	Embeddings from language models are good learners for single-cell data analysis DOI: 10.1016/j.patter.2025.101431	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-043	Technical and legal aspects of federated learning in bioinformatics: applications, challenges and opportunities DOI: 10.3389/fdgth.2025.1644291	Reviewer 1: Mention only Reviewer 2: Extend or develop	Mention only User-adjudicated consensus
SCGEN-HV-044	Contrastive Mixture of Posteriors for Counterfactual Inference, Data Integration and Fairness DOI: 10.48550/arxiv.2106.08161	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-045	scCompressSA: dual-channel self-attention based deep autoencoder model for single-cell clustering by compressing gene–gene interactions DOI: 10.1186/s12864-024-10286-2	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-046	Biological network-inspired interpretable variational autoencoder DOI: 10.1101/2020.12.17.423310	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-047	Clifti-GPT: privacy-preserving federated fine-tuning and transferable inference of foundation models on clinical single-cell data DOI: 10.1186/s13040-026-00582-w	Reviewer 1: Apply to biological data Reviewer 2: Apply to biological data	Apply to biological data Reviewer agreement
SCGEN-HV-048	D-SPIN constructs gene regulatory network models from multiplexed scRNA-seq data revealing organizing principles of cellular perturbation response DOI: 10.1101/2023.04.19.537364	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-049	Synthetic single cell RNA sequencing data from small pilot studies using deep generative models DOI: 10.1038/s41598-021-88875-4	Reviewer 1: Mention only Reviewer 2: Mention only	Mention only Reviewer agreement
SCGEN-HV-050	Deep Learning Based Models for Preimplantation Mouse and Human Development DOI: 10.1101/2024.02.16.580649	Reviewer 1: Apply to biological data Reviewer 2: Benchmark	Benchmark User-adjudicated consensus